Chapter 10: Vector Autoregression (VAR): Multivariable, multi-equation models

- Up to Sims (1980) seminal paper, almost all work done with macroeconomic modeling was carried using simultaneous equations(2SLS, 3SLS, FIML) system (Cowles Commission Method).

- Main article: Sims(1980) – "Macroeconomics and Reality"

  - Sims' work criticize the Cowles Commission's approach to large-scale structural econometric modeling: too many parameters were unidentified.

  - Sims major critic was that we should no separate variables into endogeneous and exogeneous; all variables should be treated as endogeneous one.

  - According to Sims, exclusion restrictions are ad hoc and should not be imposed. Macro models should be based on VAR models.

  - Sims' idea: estimate a model with minimal parametric restrictions and test economic theory without rely on elaborated parametrized dynamic general equilibrium models.

Why did VAR models become so popular?

- Due the oil price shock (1973), macroeconomists had difficult to make good forecast.
- VAR models changed the focus from assessing “anticipated changes in policies” to analyzing the impact of “unanticipated changes” (shocks). Assessing the effects of unanticipated shock was the real appeling for using VAR methodology.

- However, even VAR models requires identification that is usually achieved through exclusion restriction (specially Structural VAR);
- VAR: unrestricted model: allows for forecasting several variables with a single model.

- Structural VAR: it is a dynamic simultaneous equation model; theoretical identifying restrictions are imposed to unrestricted VAR. It is good for describing the data as well as for prescribing policy.

- Structural VAR includes current values of the policy variables showing up on both sides of the equal sign.

The coefficients of the unrestricted VAR are estimated by OLS. Under the VAR assumptions, the OLS estimators are consistent and have a joint normal distributions in large samples. Also, we can test joint hypothesis that involve restrictions across multiple equations.

Once the VAR model is estimate, we can graphically describe complex interactions, and we can make statements about causality and even understand something about how the structure of the real economy works.

To estimate VAR we need to use SUR regression (seemingly unrelated regression) due the fact that the errors are correlated across equations.

The Simplest Possible VAR: 2 variables and 1 lag - VAR(1)

- Two variables: X and Y
- The model:

$$X_t = \alpha_x + \beta_{x,x} X_{t-1} + \beta_{y,x} Y_{t-1} + \epsilon_{x,t}$$

$$Y_t = \alpha_y + \beta_{x,y} X_{t-1} + \beta_{y,y}\, Y_{t-1} + \epsilon_{y,t}$$

The model above can be estimated by OLS, provided $\epsilon_{x,t}$ and $\epsilon_{y,t}$ are serially uncorrelated. If error are not uncorrelated across equations, the model should be estimated by SUR (seemingly unrelated regression).

VAR models can be used to graphically describe complex interactions (impulse response functions), to make statements about causality (Granger Causality test) or understand the structure of the dreal economy (Structural VAR models).

Number of lags to include in a VAR model

- Appropriate lag length selection can be critical (it affects the variance decomposition):
  - If p is too small, the model is misspecified;
  - If p is too large, degrees of freedom are wasted.

- Standard set of lag-length selection criteria (chooses the model that minimizes the criteria):

1) AIC
2) SBIC
3) HQIC (Hannan and Quinn)

As we have seen before, the AIC and BIC has a penalty on the Likelihood function.

Approach: Estimate many VAR models with different numbers of lags and compare their fit. The command VARSOC on STATA will compute it. You will need to choose a maxlag(p) (the suggestions is chose p no larger than 5 for yearly data, 8 for quarterly data; and 15 for montly data).

Stata:

varsoc GNPgr Migr if time<yq(2017,3), maxlag(8)

SBIC: 1 lag

AIC and HQIC: 2lags

AIC is better when VAR has many equations; otherwise SBIC and HQIC perform better.

```
drop _all
freduse GNPC96 MANMM101USQ189S
gen time=yq(year(daten),  quarter(daten))
tsset time,  quarterly
gen GNPgr  =  ln(GNPC96)  -  ln(L.GNPC96)
gen M1gr  =  ln(MANMM)  -  ln(L.MANMM)
label var GNPgr  "Growth rate of real GNP"
label var M1gr  "Growth rate of M1"
drop GNPC MANMM date daten
drop if year(time)<1960
tsline GNPgr M1gr, lwidth(thick thin)
                *graph export VARdata01.eps, replace
                 graph export VARdata01.pdf, replace
reg GNPgr L.GNPgr L.M1gr        if time<yq(2017,3)
reg M1gr L.GNPgr L.M1gr         if time<yq(2017,3)
sureg (GNPgr L.GNPgr L.M1gr) (M1gr L.GNPgr L.M1gr)
var GNPgr M1gr if time<yq(2017,3), lags(1)
varbasic GNPgr M1gr if time<yq(2017,3), lags(1)
*irf graph irf, saving("IRF01")
irf graph irf
                *graph export IRF01.eps, replace
                *graph export IRF01.pdf, replace
varsoc GNPgr M1gr if time<yq(2017,3), maxlag(8)
```

Expressing VAR in matrix form

$$X_t = \beta_{1,1} X_{t-1} + \beta_{1,2} Z_{t-1} + \epsilon_{1,t}$$

$$Z_t = \beta_{2,1} X_{t-1} + \beta_{2,2} Z_{t-1} + \epsilon_{2,t}$$

$$\boldsymbol{Y_t} = \begin{bmatrix} X_t \\ Z_t \end{bmatrix}$$

Where Y_t is variable matrix and X_t and Z_t are vectors

Expressing VAR in matrix form

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix}$$

$$\boldsymbol{\epsilon_t} = \begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \end{bmatrix}$$

Where $\boldsymbol{\beta}$ is coefficient matrix and ϵ_t is the error matrix.

Expressing VAR in matrix form

$$\begin{bmatrix} X_t \\ Z_t \end{bmatrix} = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix} * \begin{bmatrix} X_{t-1} \\ Z_{t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \end{bmatrix}$$

or

$$\boldsymbol{Y_t} = \boldsymbol{\beta Y_{t-1}} + \boldsymbol{\epsilon_t}$$

is a VAR(1) model

Obs: With the appropriately define companion matrix, any VAR(p) can be rewritten as a VAR(1): the companion matrix will be useful in examining the stability of the the VAR model.

STABILITY

Method 1: Consider the univariate AR(1) case:

$$Y_t = \beta Y_{t-1} + \epsilon_t$$

The model is said to be stable if:

$$|\beta| < 1$$

Using the companion form for VAR(1) (omitting the error term), we have:

$$\boldsymbol{Y_t} = \boldsymbol{\beta Y_{t-1}}$$

This system will reaches a steady state if, as Y feeds into itself through β, it does not get bigger and bigger: $\boldsymbol{\beta}$ maps $\boldsymbol{Y_{t-1}}$ into $\boldsymbol{Y_t}$.

STABILITY

Note: Every square matrix (like $\boldsymbol{\beta}$) has at least one eigenvalue ($\boldsymbol{\nu}$) and the associated eigenvector ($\boldsymbol{\lambda}$) when:

$$\boldsymbol{\lambda\nu} = \boldsymbol{\beta\nu}$$

The vector $\boldsymbol{\nu}$ keeps its direction and only changes its magnitude when transformed by the matrix $\boldsymbol{\beta}$ (the relationships between the components stay proportional).

A stable system: the eigenvalues of matrix $\boldsymbol{\beta}$ must be less than one (or lie inside the unit circle); for complex roots, their modulus (distance) must be less than 1.

Stability: Example

- Consider the two-variable VAR(1):

$$X_t = 0.5X_{t-1} + 0.6Y_{t-1} + \epsilon_{x,t}$$

$$Y_t = -0.5X_{t-1} + 0.5Y_{t-1} + \epsilon_{y,t}$$

- The companion matrix:

$$\boldsymbol{\beta} = \begin{bmatrix} 0.5 & 0.6 \\ -0.5 & 0.5 \end{bmatrix}$$

It has 2 eigenvalues: (0.5001246 + 0.54759i) and (0.5001246 − 0.54759i)

Stability: Example

- Note that roots and be real and complex. In our example, the roots are complex. In this case, the length of a complex vector is equal to the square root of the product of the vector and its complex conjugate:

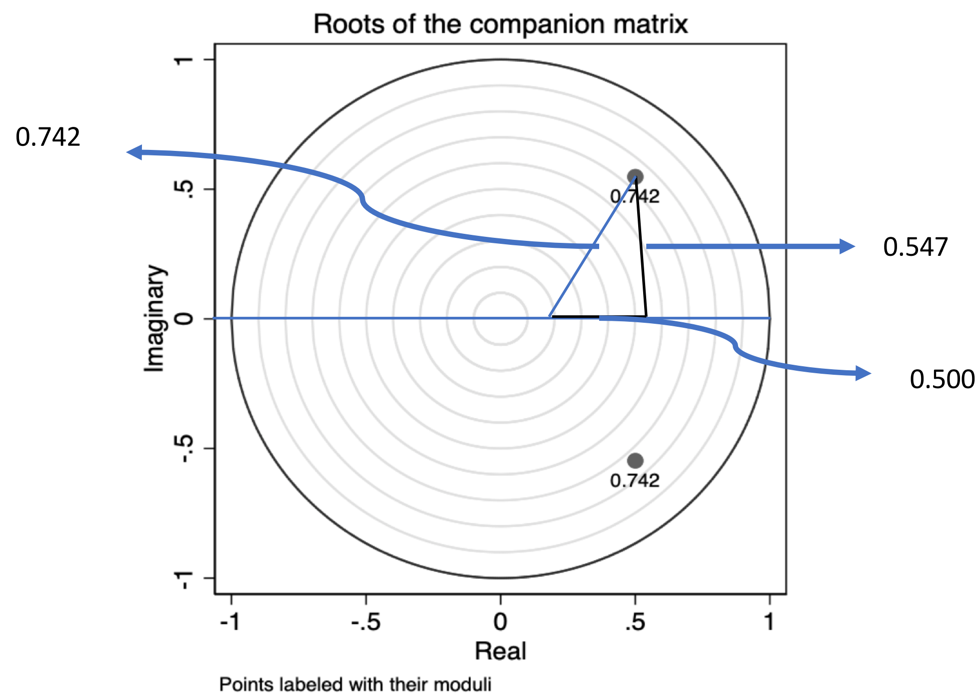
$$\sqrt{(r + c_i)(r - c_i)} = \sqrt{r^2 + c^2} = length$$

To be stable:

$$\sqrt{r^2 + c^2} < 1$$

$$\sqrt{(0.5001246)^2 + (0.54759)^2} \approx 0.742 < 1$$

This VAR model is stable (the vector needs to “shrink” to be stable)


Roots of the companion matrix
0.742
0.742
0.742
0.547
0.500
Imaginary
Real
-1
-.5
0
.5
1
Points labeled with their moduli

STABILITY

- Method 2: going back to our AR(1) case:

$$Y_t = \beta Y_{t-1} + \epsilon_t$$

$$Y_t = \beta L Y_t + \epsilon_t$$

$$Y_t - \beta L Y_t = \epsilon_t$$

$$Y_t(1 - \beta L) = \epsilon_t$$

- As we have seen before (AR models): stability requires:

$$|\beta| < 1$$

For an AR(p) the stability requires that all of the polynomial in L being greater than one (or the inverse of the characteristic equation being less than one). We can now consider the simple two-variable VAR(1) model (in the companion form):

$$\boldsymbol{Y_t} = \boldsymbol{\beta Y_{t-1}} + \boldsymbol{\epsilon_t}$$

Where:

$$\begin{bmatrix} X_t \\ Z_t \end{bmatrix} = \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix} * \begin{bmatrix} X_{t-1} \\ Z_{t-1} \end{bmatrix} + \begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \end{bmatrix}$$

- Applying the lag operator:

$$\boldsymbol{Y_t} = \boldsymbol{\beta} L \boldsymbol{Y_t} + \boldsymbol{\epsilon_t}$$

$$\boldsymbol{Y_t} - \boldsymbol{\beta} L \boldsymbol{Y_t} = \boldsymbol{\epsilon_t}$$

$$\boldsymbol{Y_t}(\mathbf{1} - \boldsymbol{\beta} L) = \boldsymbol{\epsilon_t}$$

- Solving the roots of the characteristic equation (replacing L with z):

$$\left| I - \widehat{\boldsymbol{\beta}} z \right| = 0$$

⇒ the determinant of the matrix has to be equal to zero

Going back to our previous example:

$$\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 0.5 & 0.6 \\ -0.5 & 0.5 \end{bmatrix}$$

$$0 = \left| \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.5 & 0.6 \\ -0.5 & 0.5 \end{bmatrix} z \right|$$

$$0 = \left| \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0.5z & 0.6z \\ -0.5z & 0.5z \end{bmatrix} \right|$$

$$0 = \begin{vmatrix} 1 - 0.5z & -0.6z \\ 0.5z & 1 - 0.5z \end{vmatrix}$$

- Calculating determinant of a 2x2 matrix, we have:

$$(1 - 0.5z)^2 - (0.5z)(-0.6z) = 0$$

The solution of this quadratic equation will yields a complex root:

(0.90909 + 0.9959i) and (0.90909 + 0.9959i)

- Their length is equal to 1.3484, greater than one (the eigenvalues in method 1 are the inverse of method 2 (1/1.3484 = 0.742).

- In STATA: after estimate the VAR model, use the command:

  varstable, graph

  (It will estimate the eigenvalues from method 1)

Long-Run Levels (including a constant)

- As we having seen in the AR(1) models, including a constant affect the mean of the process but it does not affect whether the process is stable.

- In a VAR model, the mean of the process is a complicated function of the constant and the other coefficients. Considering a general VAR(p) in matrix form, we have:

$$\boldsymbol{Y_t} = \boldsymbol{\beta_0} + \boldsymbol{\beta_1} \boldsymbol{Y_{t-1}} + \boldsymbol{\epsilon_t}$$

Computing the unconditional mean of this equation, known that for a stable process we have:

$$E[\boldsymbol{Y_t}] = E[\boldsymbol{Y_{t-1}}] = \boldsymbol{Y}^*; \text{ and } E[\boldsymbol{\epsilon_t}] = 0$$

The long-run mean of the process will be:

$$\boldsymbol{Y}^* = \boldsymbol{\beta_0} + \boldsymbol{\beta_1} \boldsymbol{Y}^*$$

$$\boldsymbol{Y}^* = \boldsymbol{\beta_0} [\boldsymbol{I} - \boldsymbol{\beta_1}]^{-\mathbf{1}}$$

For Y to be stable, the matrix $[\boldsymbol{I} - \boldsymbol{\beta_1}]$ has to be invertible. This condition does not depend on $\boldsymbol{\beta_0}$.

Expressing VAR as a VMA process

- We have seen that an AR(1) can be express as an MA(∞) process.
- The same can be said about the VAR(1) in its matrix form:

$$\boldsymbol{Y_t} = \boldsymbol{\beta Y_{t-1}} + \boldsymbol{\epsilon_t}$$

- Can be express as:

$$\boldsymbol{Y_t} = \sum_{j=0}^{\infty} \boldsymbol{\beta}^j \boldsymbol{\epsilon_{t-j}}$$

where: $\boldsymbol{\beta}^0 = \boldsymbol{I}$

- The weights $\boldsymbol{\beta}^j$ can be thought of as the values of the Impulse Response to a shock ($\boldsymbol{\epsilon_{t-j}}$)

Impulse Response Functions

- Impulse Response Functions (IRFs): how changes in each of the variables affect the other variables over time.

- Calculating the first values of an IRF by hand:

Example: VAR(1):

$$X_t = 0.40X_{t-1} + 0.10Y_{t-1} + \epsilon_{x,t}$$

$$Y_t = 0.20X_{t-1} - 0.10Y_{t-1} + \epsilon_{y,t}$$

Assuming: $X_0 = 0, Y_0 = 0$

Question: What will happen to X_t and Y_t if there is a one-time one-unit shock to $\epsilon_{x,t}$ (keeping all other ϵs equal to zero)?

In period 1, we will have:

$$\widehat{X_1} = 0.40(0) + 0.10(0) + 1 = 1$$

$$\widehat{Y_1} = 0.20(0) - 0.10(0) + 0 = 0$$

In period 2:

$$\widehat{X_2} = 0.40X_1 + 0.10Y_1 + 0 = 0.4$$

$$\widehat{Y_2} = 0.20X_1 - 0.10Y_1 + 0 = 0.2$$

In period 3:

$$\widehat{X_3} = 0.40X_2 + 0.10Y_2 + 0 = 0.18$$

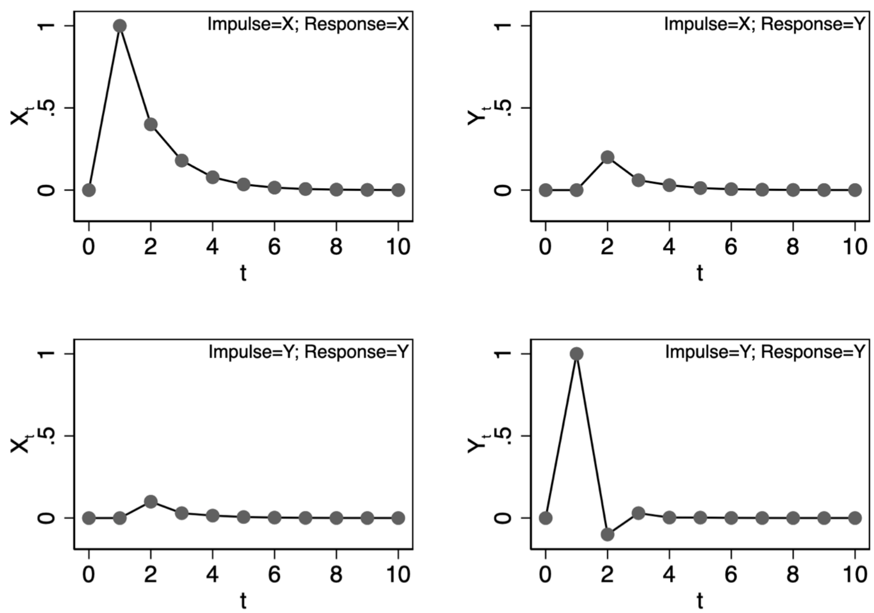
$$\widehat{Y_3} = 0.20X_2 - 0.10Y_2 + 0 = 0.06$$

In period 4:

$$\widehat{X_4} = 0.40X_3 + 0.10Y_3 + 0 = 0.078$$

$$\widehat{Y_4} = 0.20X_3 - 0.10Y_3 + 0 = 0.0.30$$

We calculate the impulse responses from X to X and X to Y



Using the MA(∞) representation of a stationary VAR(1) to compute the impulse response function

$$\boldsymbol{Y_t} = \sum_{j=0}^{\infty} \boldsymbol{\beta}^j \boldsymbol{\epsilon_{t-j}}$$

- The weights $\boldsymbol{\beta}^j$ can be thought of as the values of the impulse response to a shock to $\boldsymbol{\epsilon_{t-j}}$. The sequence of $\boldsymbol{\beta}^j$s are the IRFs.
- Going back to our previous example of a two-variable VAR(1), we have (considering only the left column entries):

$$\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 0.40 & 0.10 \\ 0.20 & -0.10 \end{bmatrix}$$

$$\widehat{\boldsymbol{\beta}}^2 = \begin{bmatrix} 0.40 & 0.10 \\ 0.20 & -0.10 \end{bmatrix} \begin{bmatrix} 0.40 & 0.10 \\ 0.20 & -0.10 \end{bmatrix} = \begin{bmatrix} 0.18 & 0.03 \\ 0.06 & 0.03 \end{bmatrix}$$

$$\widehat{\boldsymbol{\beta}}^3 = \begin{bmatrix} 0.078 & 0.015 \\ 0.03 & 0.03 \end{bmatrix}$$

FORECASTING

- VAR models are very popular due its forecasting feature. To forecast period $t+1$ we only need the past values of the variables. However, to forecast $t+k$ we will need to rely on previous period forecast.
- Example: VAR(2), two-variables (n=100):

$$X_t = 0.30X_{t-1} + 0.20X_{t-2} + 0.10Y_{t-1} + 0.05Y_{t-2} + \epsilon_{1,t}$$

$$Y_t = 0.35X_{t-1} + 0.25X_{t-2} + 0.15Y_{t-2} + 0.01Y_{t-2} + \epsilon_{2,t}$$

- Suppose we want to forecast for the next 2 periods ahead:

  Computing the conditional expectation on data up to period 100, we have:

$$E(X_{101}) = 0.30E(X_{100}) + 0.20E(X_{99}) + 0.10E(Y_{100}) + 0.05E(Y_{99}) + E\left(\epsilon_{1,101}\right)$$

$$E(Y_{101}) = 0.35E(X_{100}) + 0.25E(X_{99}) + 0.15E(Y_{100}) + 0.01E(Y_{99}) + E\left(\epsilon_{2,101}\right)$$

$$E(X_{101}) = 0.30(1.296) + 0.20(0.220) + 0.10(0.972) + 0.05(-0.54) + 0 = 0.5277111$$

$$E(Y_{101}) = 0.35(1.296) + 0.25(0.220) + 0.15(0.972) + 0.01(-0.54) + 0 = 0.6544008$$

- To forecast two periods ahead we will need to use $E(X_{101})$ and $E(Y_{101})$:

$$E(X_{102}) = 0.30(0.5277) + 0.20(1.296) + 0.10(0.654) + 0.05(0.972) + 0 = 0.53175912$$

$$E(Y_{102}) = 0.35(0.5277) + 0.25(1.296) + 0.15(0.654) + 0.01(0.972) + 0 = 0.61680386$$

Note: the far ahead you are forecasting, the less certain we are of our forecasts. Best to re-estimate the VAR and re-forecast when data are available.

STATA command fcast compute will automatically compute the forecast values (after estimating the model):

```
fcast compute E, nose  step(2)
```

Note: this command creates several new variables with a given prefix (the textbook author choose the prefix E (due to expectation), you can use any letter).

Option nose: it instruct STATA not to compute any standard errors.

Option step: tells STATA that we want # of steps ahead (forecast horizon).

Grange Causality

- If changes in X predate changes in Y, we may think that X cause changes in Y and there will be a correlation between Y and lagged X (careful, correlation is not the same thing as causation).
- A variable X is said to Granger-cause Y if including past values of X in our model helps us to predict Y (better than if we do not include it).
- The causality known as Granger causality is a necessary condition for existential causality, but I not a sufficient condition (because X could be Granger-causes Y indirectly through another variable, Z. So, Granger causality is not a strict causality.

- Example: univariate AR(2) with and exogeneous variable (X):

$$Y_t = \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \beta_3 X_{t-1} + \epsilon_t$$

The model above is said to be better, provide β_3 is statistically significantly, at predicting Y than a more restricted model that does not include the past value of X. If the model include more lags of X, then a joint F-test of all the coefficients on the lagged X variables is test (null: all the coefficients being equal to zero; alternative: at least one is different than zero)

Granger causality in a VAR model: the case of VAR(2)

$$X_t = \beta_{1,1}X_{t-1} + \beta_{1,2}Y_{t-1} + \beta_{1,3}X_{t-2} + \beta_{1,4}Y_{t-2} + \epsilon_{x,t}$$

$$Y_t = \beta_{2,1}X_{t-1} + \beta_{2,2}\, Y_{t-1} + \beta_{2,3}X_{t-2} + \beta_{2,4}Y_{t-2} + \epsilon_{y,t}$$

- If $\beta_{2,1}$ and $\beta_{2,3}$ are jointly significant from zero, we say that X Granger-cause Y.
- Note: The Granger causality tests are sensitive to omitted variables (so the importance of choosing the correct number of lags) and other variables in the VAR model and can induce a "false positive" or "false negative results to the tests: it is very important to have knowledge of the economic theories to know which relevant variables to include in the model.

Granger causality test: unemployment and inflation

Quietly var Infl Unemp if year(daten) < 1980, lag(1 2)

vargranger

Granger causality Wald tests

Equation	Excluded	chi2	df	Prob > chi2
Infl	Unemp	5.0299	2	0.081
Infl	ALL	5.0299	2	0.081
Unemp	Infl	7.6007	2	0.022
Unemp	ALL	7.6007	2	0.022

Based on the results above, unemployment does not Granger-cause inflation (0.081 > 0.05); but inflation does Granger-cause unemployment (0.022< 0.05).

Chris Sims' application to Granger's causality test

- In his Money, Income and Causality (1972) paper, Sims tested whether changes in money supply cause fluctuations in GNP, or vice versa. The results were very important to the debate Monetarism vs Keynesianism: are monetary policy effective in smoothing out the business cycle?

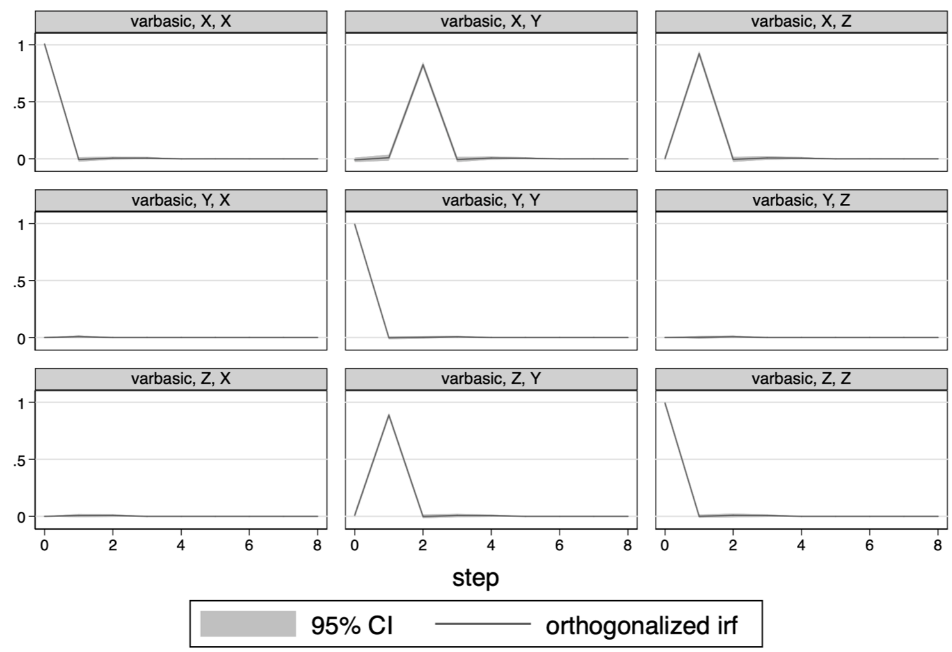
- Sims conclude that the money supply (monetary base) affects GNP, and that GNP does not affect the money supply (lags were arbitrarily chosen and Hsiao (79, 81) proposed using Akaike's FPE criteria to select different lag length for each variable and each equation). Different models gives different results: you should not choose lags arbitrarily.

- Stock and Watson(89) showed that adding a deterministic time trend strengthens money's estimated effect on output; they showed also that the sample data can affect the results. Their main economic finding: shocks to the money growth rate that are greater than those predicted by the trend do have an effect on output.

Indirect Causality

- It should be noted that Granger causality test cannot detect indirect causality: X causes Z and then Z causes Y (but X does not cause Y directly). This occurs because Granger causality tests are single equation test and indirect causality operates through multiple equations (see simulation on pages 301 and 302 showing that the Granger causality test does not show that X causes Y).
- Question: Does the indirect effect of X on Y show up in the OIRFs (orthogonalized IRFs)? Yes, they will show how changes in X (and only in X) affect all the other variables in the VAR system.
- Note: IRFs are not useful if shocks are correlated across variables (that why VAR is estimated using SUR, that allows for contemporaneously correlated errors across equations). The OIRFs allow us to measure the impact of an exogenous change in a policy variable, and track the effect on the other variables (for this we need shocks to orthogonal to each other – un-correlated).

Next page graphs of OIRFs show that a shock to X is followed by an response in Y (see top center panel):

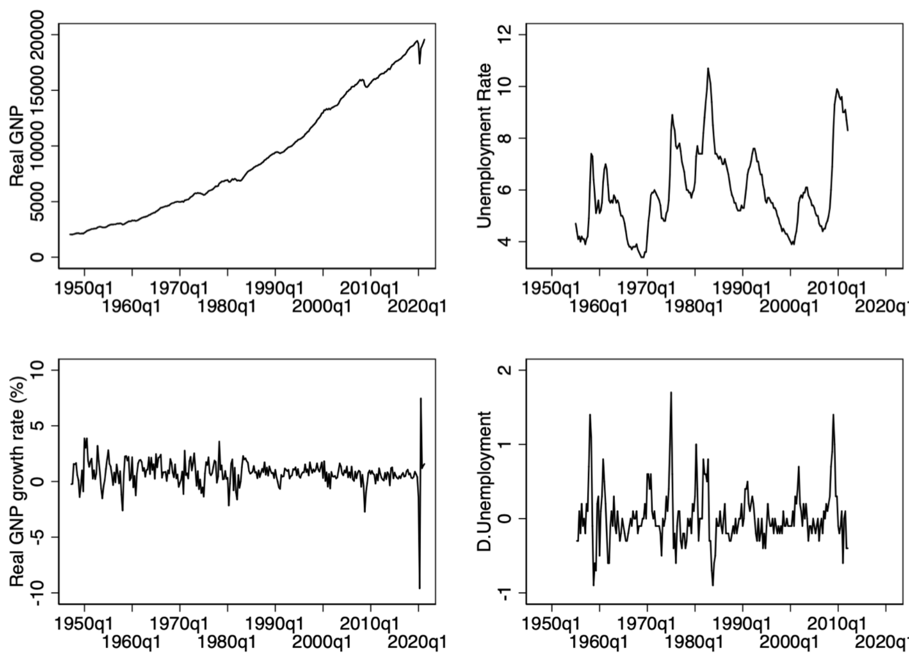


Graphs by irfname, impulse variable, and response variable

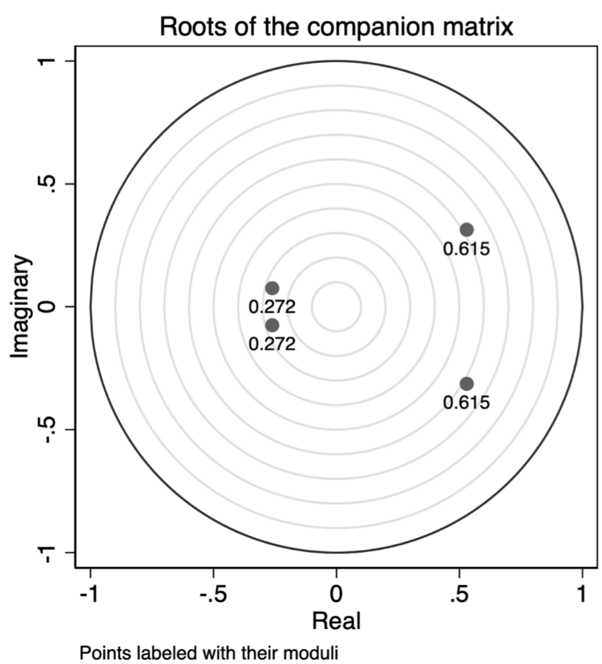
Example: GNP and Unemployment (USA)

- Steps:
  1) Make sure your data is stationary (if not, apply the appropriate transformation);
  2) Determine the number of lags;
  3) Estimate the VAR model;
  4) Check if the estimated model is stable;
  5) Test for Granger causality and calculate IRFs and FEVDs (Forecast Error Variance Decomposition)

  Forecast Error Variance Decomposition: it can tell us how important each shock is (or how much of the variation in X is due to shocks in Y), while the IRF tell us how a shock to one variable propagates through the system and affects the other variables. They also allow us to know how X affects Y across different lags.



Real GNP
0
5000
10000
15000
20000
Unemployment Rate
4
6
8
10
12
Real GNP growth rate (%)
-10
-5
0
5
10
D.Unemployment
-1
0
1
2
1950q1
1960q1
1970q1
1980q1
1990q1
2000q1
2010q1
2020q1



Roots of the companion matrix
Imaginary
1
.5
0
-.5
-1
-1
-.5
0
.5
1
Real
0.615
0.272
0.272
0.615
Points labeled with their moduli

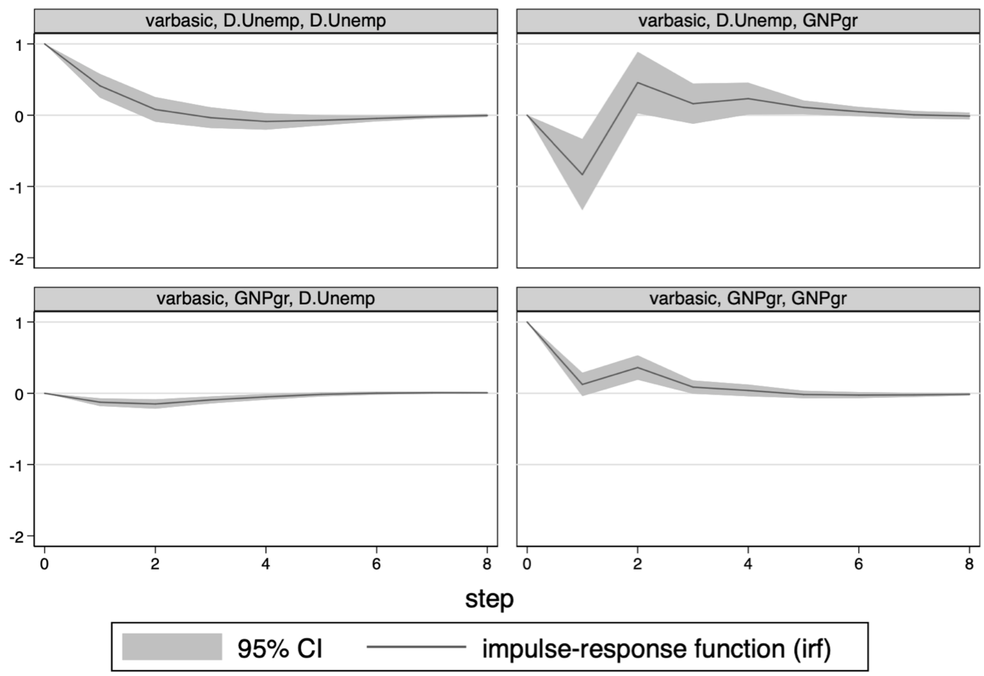
Granger Causality test

- Both growth rate of GNP Granger causes the first difference of unemployment (first difference is used because we could not reject the null of unit roots) and the first difference of unemployment Granger cause growth rate of GNP.
- The IRFs and FEVDs help us to understand the dynamics between these two variables:

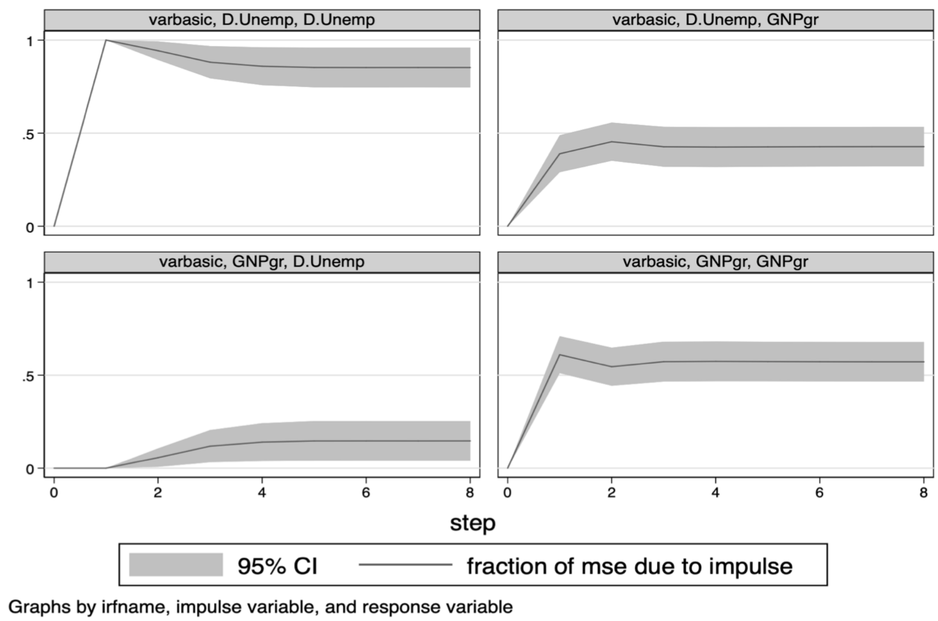
IRFs results:

- Shocks to D.Unemp dampens gently to zero by the forth quarter and it decreases the GNP growth rate for a quarter, before it returns to its baseline level.
- Shocks to the GNP growth rate decrease the unemployment growth rate only slightly

- FEVDs results:



Graphs by irfname, impulse variable, and response variable



Forecast error variance decomposition

Structural VAR (SVAR)

- Critics to VAR approach: devoid of any economic content.
- Question: How does it differ from the "reduced form VAR"?
- Structural VAR: uses economic theory, rather than the Choleski decomposition, to recover the structural innovation from the residual of the reduced form.
- Reduced VAR is bet suited for data description but it is will suited for policy prescription. Structural VAR will allow us to forecast what will happen to the variables if the structure of the economy changes.
- A structural VAR has contemporaneous terms (present values) of X_t and Y_t as dependent variables in each equation. Reduced VAR, as we have seen, have only lagged values.
- Problem with SVAR models: IDENTIFICATION.

Example: SVAR(1):

$$X_t = \alpha_{y,o} Y_t + \alpha_{x,x} X_{t-1} + \alpha_{x,y} Y_{t-1} + e_{x,t}$$

$$Y_t = \alpha_{x,o} X_t + \alpha_{y,x} X_{t-1} + \alpha_{y,y} Y_{t-1} + e_{y,t}$$

- In matrix notation we will have:

$$\boldsymbol{A_0 Y_t} = \boldsymbol{A_1 Y_{t-1}} + \boldsymbol{e_t}$$

Where:

$\boldsymbol{A_0} = \begin{bmatrix} 1 & \alpha_{y,o} \\ \alpha_{x,o} & 1 \end{bmatrix}$, $\boldsymbol{A_1} = \begin{bmatrix} \alpha_{y,x} & \alpha_{x,y} \\ \alpha_{y,x} & \alpha_{y,y} \end{bmatrix}$, and $\boldsymbol{e_t} = \begin{bmatrix} e_{x,t} \\ e_{y,t} \end{bmatrix}$

Reduced Form of the model: SVAR(1)

$$\boldsymbol{Y_t} = \boldsymbol{\beta Y_{t-1}} + \boldsymbol{\epsilon_t}$$

$$\boldsymbol{Y_t} - \boldsymbol{\beta Y_{t-1}} = \boldsymbol{\epsilon_t}$$

$$(\mathbf{1} - \boldsymbol{\beta L})\boldsymbol{Y_t} = \boldsymbol{\epsilon_t}$$

Where:

$\boldsymbol{\beta} = \boldsymbol{A_0}^{-1} * \boldsymbol{A_1}$, and $\boldsymbol{\varepsilon_t} = \boldsymbol{A_0}^{-1} * \boldsymbol{\epsilon_t}$

where: $\boldsymbol{Y_t}$ is a $n_Y \times 1$ vector of variables. All endogeneous ones.
$\boldsymbol{\epsilon_t}$ is a $n_\epsilon \times 1$ vector of shocks. It incorporates exogeneous shocks, like technology shocks or unexpected inflation.

- If we assume the model is stable, then we can write the model above as a SVMA.

Identification of Structural VAR

- Important: the parameters of the SVAR cannot be estimated (due to the feedback inherent in the process); the reduced form of it can be estimated but cannot be used to find the SVAR parameters (there will be infinite combinations of ${\boldsymbol{A_0}}^{-1}$ and $\boldsymbol{A_1}$ that gives $\boldsymbol{\beta}$).
- Identification requires $\frac{(n^2-n)}{2}$ restrictions, in a n-variable VAR, on the parameters of the SVAR equations.
- In order to estimate a structural VAR we need the so-called "identifying restrictions", so we can solve our system of equations (going from the reduced form to the structural one) ⇒ "backing out"

Identification of Structural VAR

- How to determine the “identifying restrictions”? ⇒ ECONOMIC THEORY
- SVAR Models reintroduce economic theory into the VAR methodology (no longer the data can speak for themselves).
- In order to estimate a structural VAR we need the so-called “identifying restrictions, so we can solve our system of equations (going from the reduced form to the structural one) ⇒ “backing out”
- Structural VAR models have been identified by by imposing restrictions on the covariance matrix of the structural shocks, the matrix of contemporaneous coefficients ($\boldsymbol{A_0}$) and the matrix of long-run multipliers(restrictions on the sum of the impulse response).

The Structural VAR: the moving average model: VMA

- We have seen that we can be express a VAR(p) model as an VMA(∞) process. The same for the SVAR

$$(1 - \boldsymbol{\beta} L)\boldsymbol{Y_t} = \boldsymbol{\epsilon_t}$$

$$\boldsymbol{Y_t} = C(L)\boldsymbol{\epsilon_t}$$

Where: $C(L) = (1 - \boldsymbol{\beta} L)^{-1}$

$C(L)$ is a matrix of lagged polynomial. $C(L) = \sum_{j=0}^{\infty} C_j L^j$. It describes the long-run responses to the structural shocks.

The Impulse Response Function

- Question: how the system's endogeneous variables respond dynamically to the exogeneous shocks ($\boldsymbol{\epsilon_t}$ represents innovations or unanticipated shifts in the exogeneous variables)?

- The VMA representation of the VAR model allow us to trace out the time path of the various shocks on the variables contained on the system.

- $\frac{\partial y}{\partial \epsilon_t'} = C_o, \quad \frac{\partial y}{\partial \epsilon_{t-1}'} = C_1, \quad \frac{\partial y}{\partial \epsilon_{t-2}'} = C_2, \ \dots \dots$

- C_o is the impact effect of a unit change in ϵ_t on the endogeneous variables;
- C_1 is the impact effect of a unit change in ϵ_t on the endogeneous variables one period later;
- C_2 is the impact effect of ϵ_t on the endogeneous variables two periods later
  - We can compute the impulse response function of each $\epsilon_{j,t}$ on $Y_{i,t}$

- The accumulated effects of a unit impulse can be obtained by the summation of the coefficient C_i . As n tends to infinite, this summation will yield the long-run multiplier effect. When both variables are stationary, this summation is finite.

The Structural VAR(p) representation of VMA

- The VMA cannot be estimated due the infinite number of parameters. We will need a parametric structure that involves few parameters (like the ARMA models in the univariate case).
- Connections between VAR and VMA:

$$A(L)\boldsymbol{Y_t} = \boldsymbol{\epsilon_t}$$

Where $A(L) = C(L)^{-1}$

- The invertibility requirements:
  - $n_Y = n_\epsilon$
  - $|C(z)|$ has all of its roots outside the unit circle.

Non-invertibility in a model results in a reduction in the information set available to the econometrician, because those models will have the same second moment properties as models with roots outside and will lead to large specification errors that cannot be detected from the data.

Variance Decomposition

- The impulse response function tell us how a shock to one variable propagates through the system and affects the variables.
- Question: How important each shock is?

or

- How much of the variation in X_t is due to shocks in Y_t? Or, what percent of the variation in X from its forecast value is due to shocks directly affecting X?
- To answer this question we need to compute:

Forecast Error Variance Decomposition: FEVD

Variance Decomposition

- Our interest also is measuring which shock are the primary causes of variability in the endogeneous variables. This will help us to know which shock is the major contributor to business cycle fluctuations (variability).
- Because shock are unexpected, we can measure the relative importance of it in terms of the h-step-ahead forecast error of Y_t.
- Whenever we forecast our model, there will be some forecast error (the residual): the difference between what we expect Y to be after a certain number of periods, given the parameter estimated in our model, and the actual value of Y in that period.
- The variance decomposition splits up the variance of this residual into its component causes, and express the results as a %.

- It is possible to decompose the n-step-ahead forecast error variance into the proportions due to each shock. This decomposing will tell us the proportion of the movements in a sequence due to its own shocks versus shocks to the other variable.
- Findings on applied research: a variable normally explain almost all of its forecast error variance at short horizons and very small proportion at long horizons.
- The fraction of the variance in $\boldsymbol{e_{t|t-h}}$ that is explained by the shock will give us an idea about the importance of the shock.
- When shocks are uncorrelated, the variance decomposition is easly computed. However, when the shocks are mutually correlated, there is no unique way to compute it (necessity of the orthogonalization).

Innovation Accounting

- Impulse analysis and variance decomposition are called: Innovation Accounting. It is normally used to examine the relationship among economic variables.

Long-Run Restriction: Blanchard-Quah(1989)

- Paper: The Dynamic Effects of Aggregate Demand and Supply Disturbances; Blanchard-Quah (1989), American Economic Review.
- Their goal is to know the dynamic effects on output and unemployment after shocks.

- Motivation: Macro debate about output movements:

  - Business cycle (the dynamics of actual output around its trend)   versus

  - “Trend”(part of output that would realize were all prices perfectly flexible.

Long-Run Restriction: Blanchard-Quah(1989)

- traditional Keynesian view of fluctuations.
  - Due to nominal rigidities, demand disturbances have short-run effects on output and unemployment (Keynes assumed that prices and wages are Sticky in the short-run), but those effects disappear over time.
  - In the long run, only supply (productivity) disturbances affect output.
  - Neither disturbances have long-run impact on unemployment.
- Two types of disturbances (assumed to be serially uncorrelated with each other and neither has a long-run effect on unemployment):
  - Demand disturbances: have a hump-shaped mirrow-image effect on output and unemployment (unemployment shocks has zero long-run impact on GNP growth rate);
  - Supply disturbances: output increases steadily over time, peaking after 2 years and reaching a plateau after five years.

Long-Run Restriction: Blanchard-Quah(1989)

- Their results are consistent with a traditional view of the dynamic effects of aggregate demand on output and unemployment, in which movements in aggregate demand build up until the adjustment of prices and wages leads to the economy back to equilibrium.

Long-Run Restriction: Blanchard-Quah(1989)

- Matters: their choice of identifying restrictions. Restrictions are imposed on the sum of the impulse responses Blanchard and Quah argue that an unemployment shock has zero long-run impact on GNP growth rate:

$$\boldsymbol{Y_t} = C(L)\boldsymbol{\epsilon_t}$$

$$\begin{bmatrix} YGR_t \\ UNEMP_t \end{bmatrix} = \begin{bmatrix} c_{1,1} & c_{1,2} \\ c_{2,1} & c_{2,2} \end{bmatrix} \begin{bmatrix} \boldsymbol{\epsilon_{y,t}} \\ \boldsymbol{\epsilon_{u,t}} \end{bmatrix}$$

- They set $c_{1,2}$ equal to zero (restriction on the C matrix) ⇒

$$YGR_t = c_{1,1}\, \boldsymbol{\epsilon_{y,t}}$$

$$UNEMP_t = c_{2,1}\, \boldsymbol{\epsilon_{y,t}} + c_{2,2}\, \boldsymbol{\epsilon_{u,t}}$$

Steps to use Blanchard and Quah approach

- Step 1:
  - Pretest the variables for unit roots;
  - Perform lag-length tests to find a reasonable specification for VAR.
- Step 2:
  - Using the residuals of the estimated VAR, calculate the var/cov matrix to be able to obtain the impulse response functions and the variance decomposition.

  Blanchard and Quah approach allow structural innovations to be identify by looking at the accumulated effects of shocks.
- Problem with this technique: how to identify many types of distinctive shocks?
- Also, Blanchard and Quah take some ad-hoc steps to ensure their data is stationary (check the STATA commands on Blanchard and Quah estimation). They should have done UR tests with breaks.